FOR POWER SUPPLIES

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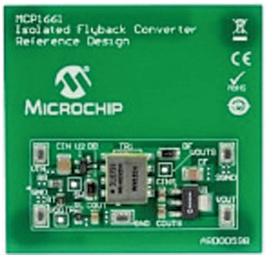
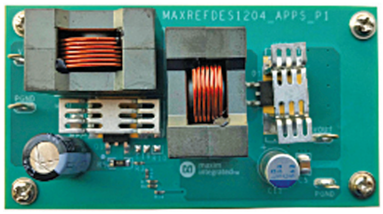
Reference Design for an Isolated Flyback Converter	Reference Design for a Single-Output SEPIC Converter	Reference Design for a 35W Wide Input Range Flyback Converter
		
Reference design for a DC-DC converter which employs a flyback topology and provides galvanic isolation, thus making it suitable for sensitive applications.	Reference design for a single-output SEPIC converter which can operate over a wide input range of 6V to 18V. It offers an efficiency of over 90% and a voltage ripple lower than 0.7%.	Reference design for a 35W flyback converter that can take a wide range of AC inputs from 90 to 305Vrms to provide a constant output of 48V.
The MCP1661 Flyback Converter Reference Design from Microchip is an isolated flyback converter that offers galvanic isolation. The MCP1661 is a constant pulse-width modulation (PWM) frequency boost (step-up) converter based on a peak current mode architecture, which delivers high efficiency over a wide load range from two-cell and three-cell alkaline, energizer ultimate lithium, NiMH, NiC, and single-cell Li-ion battery inputs. It can take inputs from 4.75V to 5.25V and is suitable for applications that require higher output voltage capabilities. It can regulate the output voltage up to 32V and deliver 125mA typical load at 3.3V input and 12V output. The power supply offers a high efficiency of up to 92%. At light loads, MCP1661 skips pulse to keep the output voltage in regulation, but the voltage ripple is maintained low. The high level of integration of the design lowers total system cost, eases implementation, and reduces board area.	The MAXREFDES1204 from Analog Devices is a reference design for a single-output SEPIC converter based on the MAX16990 controller. This reference design is rated to operate over a wide input voltage range of 6V to 18V and deliver an output power of up to 24W. The MAXREFDES1204 reference design offers a high efficiency of up to 90% while having an output ripple voltage lower than 0.7% at nominal input voltages. The power supply is capable of delivering a maximum current of 2A. The MAXREFDES1204 works in continuous current mode (CCM) and hence the output voltage depends just on the PWM. Therefore, the design is ideal for constant loads with less input noise and less output ripple spectrum but more attention to fast reverse recovery diode losses. It is suitable for use in harsh automotive environments and can be used as a power supply for automotive LED lighting, audio systems, navigation systems, dashboard equipment, etc.	The STEVAL-ILL069V2 is a reference design of a power supply based on flyback topology. It can take input directly from AC mains and provide a stable DC output of 48V. The reference design can work from 90 to 305V AC, running at a frequency of 45 to 66Hz. The design can provide a maximum power of 35W with an efficiency of up to 90%. An auxiliary 16V output is present to supply small circuits that absorb a maximum current of 20mA. The reference design has a compact size measuring just 130mm x 50mm and is ROHS-compliant. The STEVAL-ILL069V2 reference design has multiple safety features, including output shortcircuit and overload protection, and comes with an auto-restart for safe operation in lighting environments. Due to its stable output voltage, the reference design is suitable for use as an LED power supply for displays and signboards, outdoor lighting, etc.
The reference design is suitable for applications such as camera phone flash, portable medical equipment, and hand-held instruments.	The reference design is suitable for powering automotive equipment which run on 12V load, such as LED lights, vehicle infotainment systems, etc.	The reference design is suitable for use in LED applications. It can directly take supply from AC mains and power LED lights, signage, etc.
Microchip	Analog Devices	STMicroelectronics
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